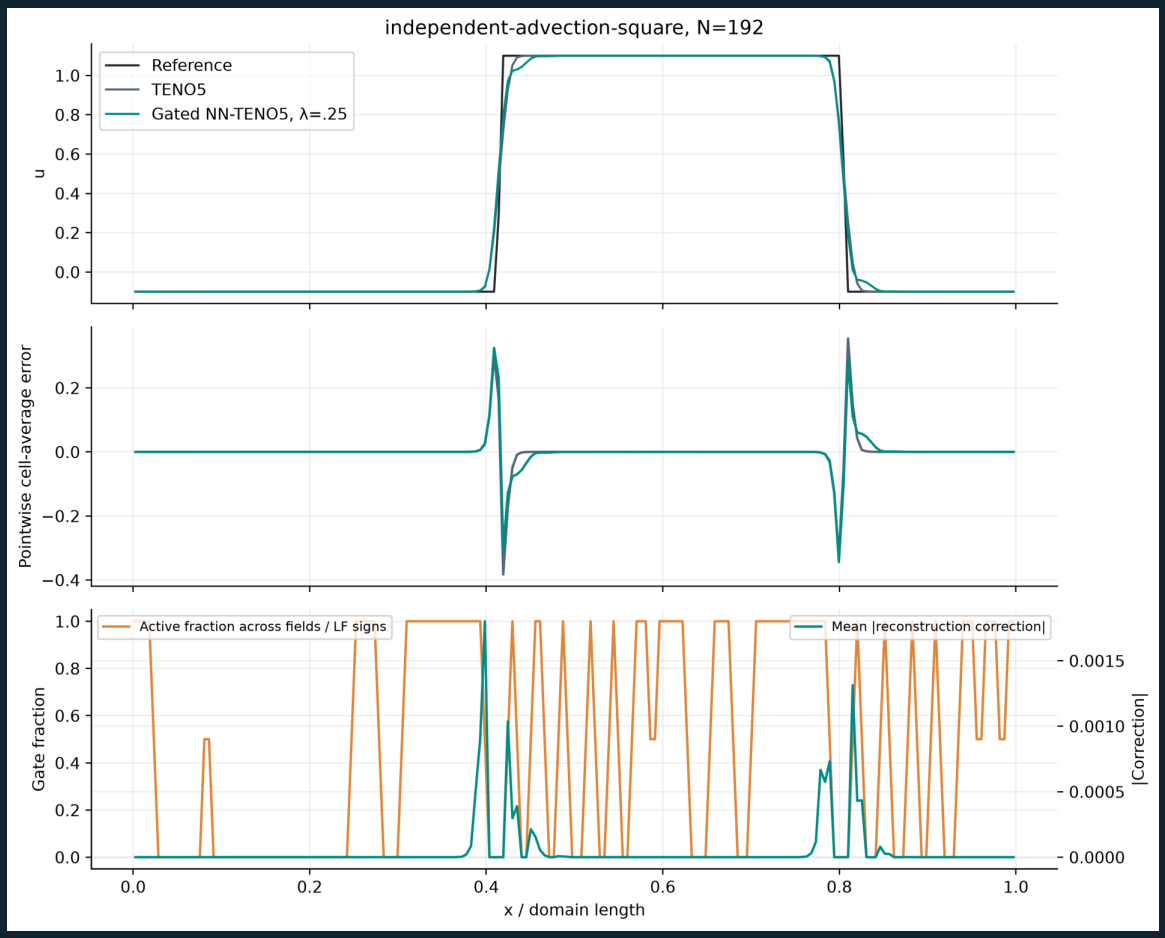
Capturing shocks accurately

A research internship in numerical PDEs and scientific machine learning.
High resolution matters; oscillations and instability matter too.

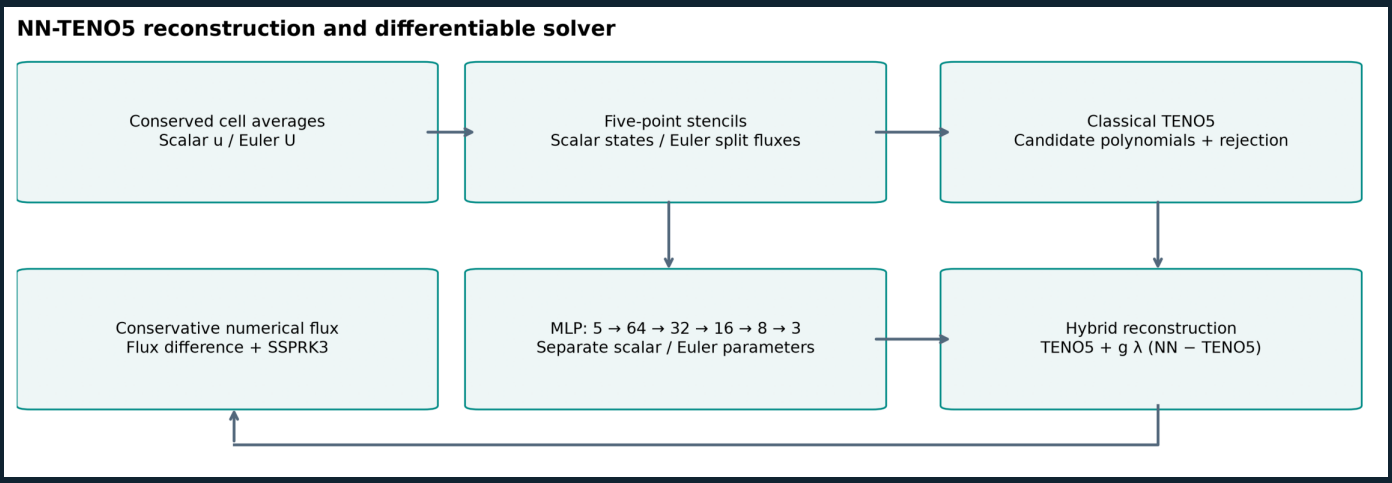


github.com/xxxxxxxx15339/Neural-Shock-Capturing

Validation-selected results are not an independent-test guarantee.

Classical structure, learned decisions

TENO5 combines candidate reconstructions. A small neural network changes their selection; the numerical flux still drives the PDE update.

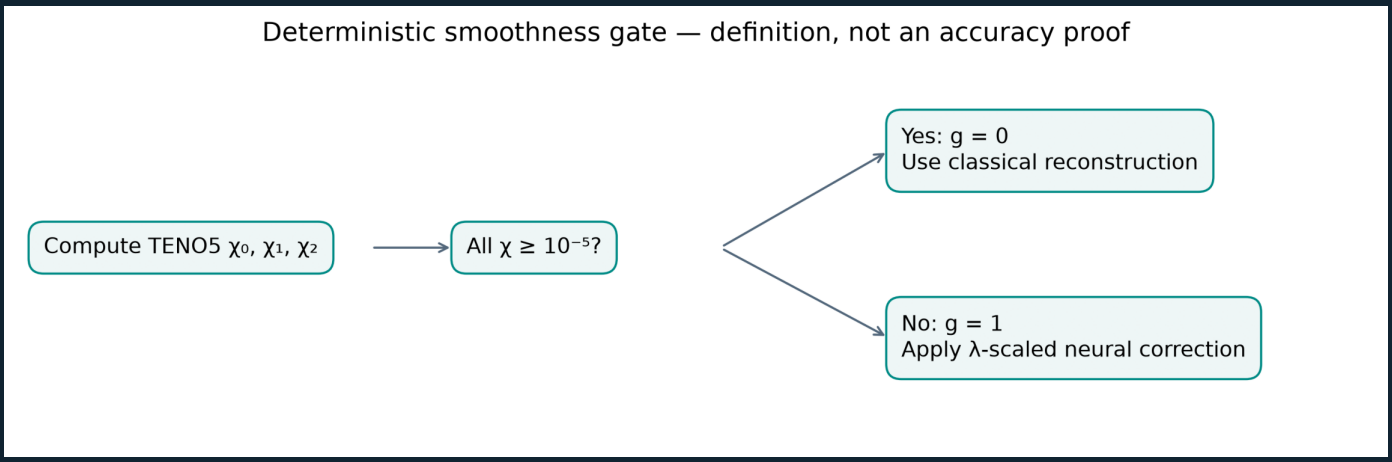


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Validation-selected results are not an independent-test guarantee.

A deterministic smoothness gate

5 → 64 → 32 → 16 → 8 → 3. Separate scalar and Euler checkpoints.
Neural corrections activate only where TENO5 rejects a candidate.

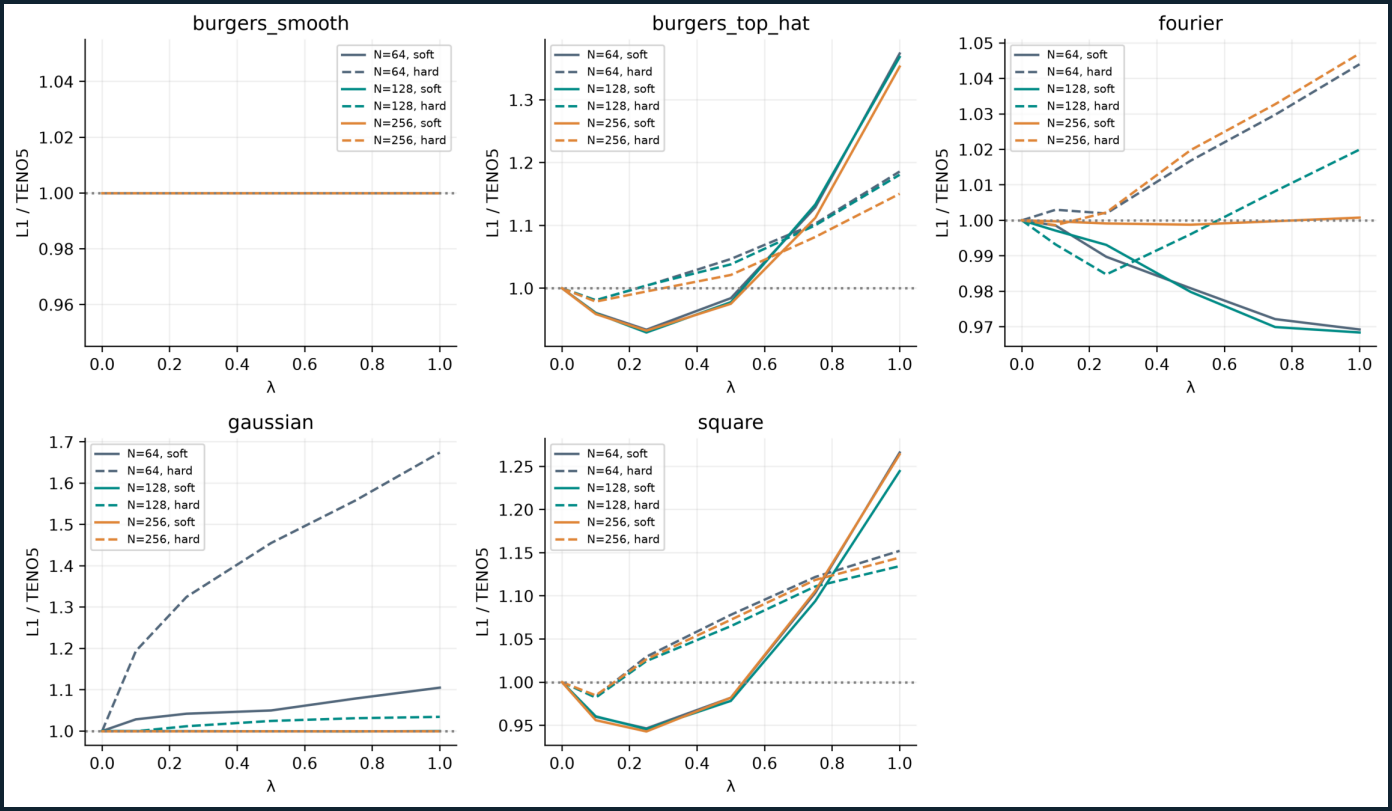


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Train against mathematical solutions

6,144 independent analytic PDE trajectories; N = 64, 128, 256.
Finite-volume cell-average targets. No experimental sensor data.

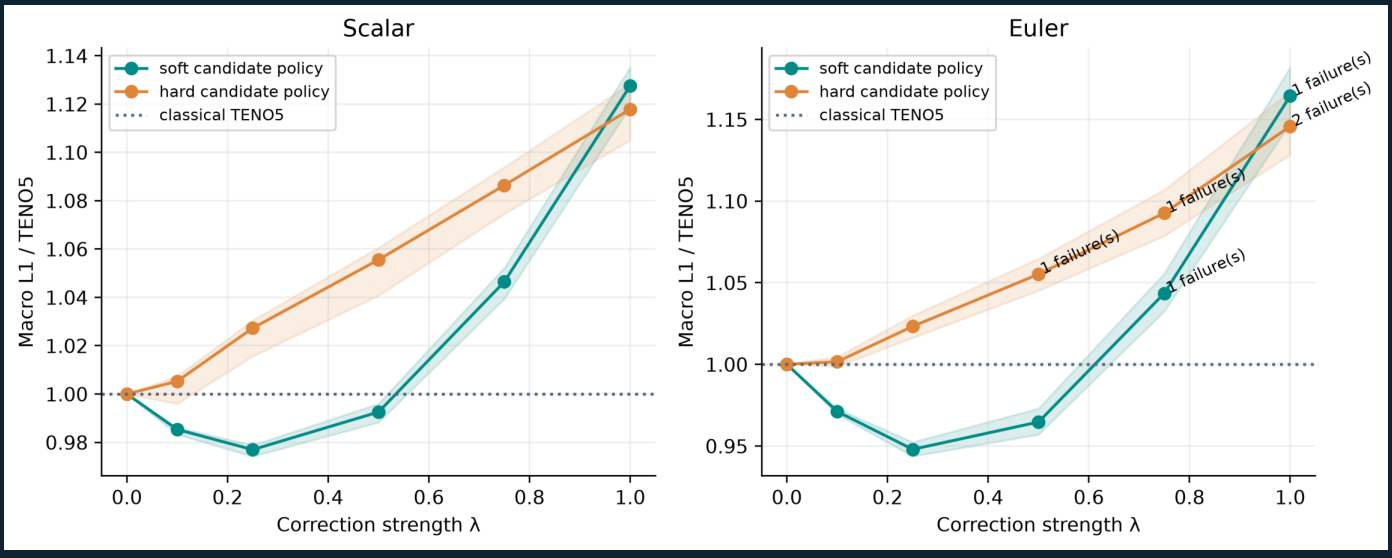


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Validation-selected results are not an independent-test guarantee.

Validation results are mixed

Full validation favors soft λ = .25: scalar macro ratio .9770; Euler .9479.
Large strengths can fail. Independent results are shown with caveats.

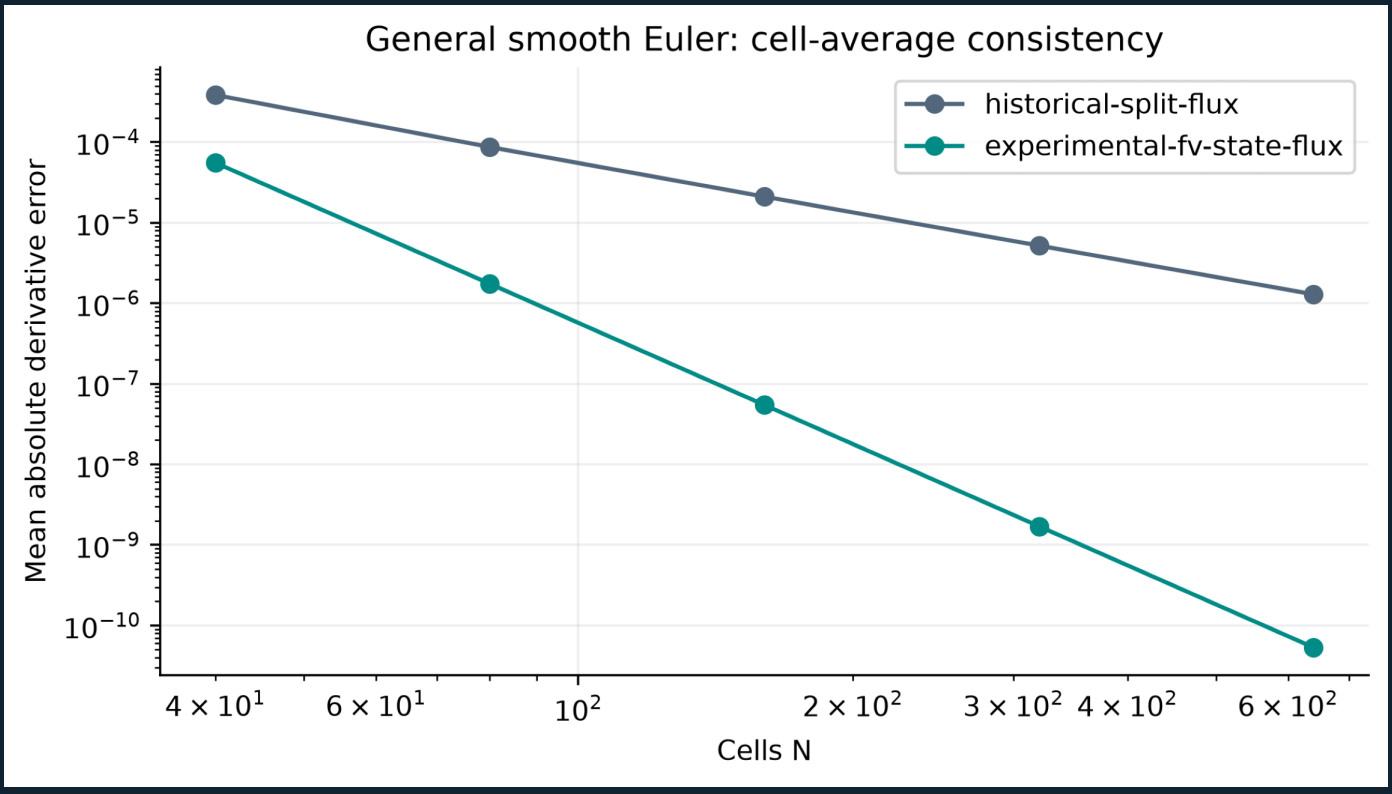


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Accuracy needs numerical evidence

Critical points and general nonlinear Euler expose accuracy limits.
Timing includes CPU inference cost; no universal speedup is claimed.



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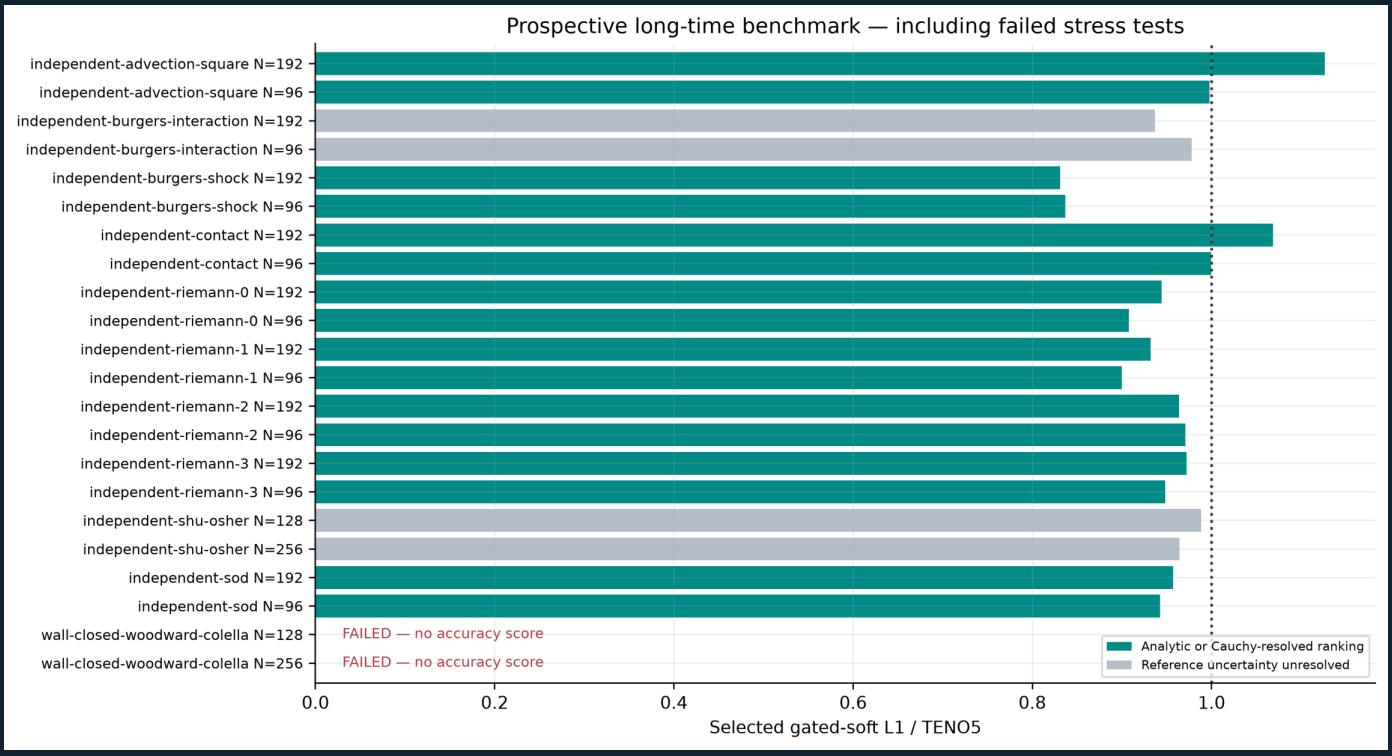
Validation-selected results are not an independent-test guarantee.

A reproducible investigation

Yasser BAOUZIL · supervision: Radouan BOUKHARFANE

Foundations: Stevens & Colonius; WENO/TENO literature.

Code, measured results, negative findings and reproduction commands.



github.com/xxxxxxxx15339/Neural-Shock-Capturing

Validation-selected results are not an independent-test guarantee.